

Estimation of voter transitions based on ecological inference: an empirical assessment of different approaches

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Abstract The analysis of voter transitions is an important area of electoral studies. A main strategy is to use aggregate data provided by the offices of statistics regarding districts, precincts, communities etc. and to rely on ecological inference. Ecological inference, however, is plagued by the well-known indeterminacy problem. In this article, we present the so far most extensive systematic empirical comparison of commonly used approaches for ecological inference of the analysis of voter transitions. Our evaluation is based on diverse simulations for multiple assumptions and scenarios. Based on recent election data for the German metropolitan city Munich, we are able to show that an application of the hierarchical multinomial-Dirichlet model, which is implemented in the R-library eiPack, exhibits the best overall estimation performance. Other prominent approaches frequently used by practitioners, e.g. the Thomsen logit approach, proved to be inconsistent. Furthermore, we demonstrate that appropriate data preprocessing is crucial for achieving reliable results.

Keywords Ecological inference · Voter transitions · Hierarchical Bayesian models

1 Introduction

The presentation of voter transitions is a highly popular part of electoral analyses on German TV in elections nights. Currently only one broadcaster, the public ARD,

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provides this cost-intensive service in collaboration with the survey company Infratest dimap who performs the complex fieldwork. Their estimates for voter transitions are exclusively provided for federal and regional elections. However, there is a growing interest for such studies at all levels (federal, regional, local) for obvious reasons: in times of rising electoral volatilities politicians are keen to know the exact patterns of gains and losses to draw conclusions regarding position-taking, coalition formation, and policy-making.

The increasing demand is confronted with major, well-known problems. Individual level data require a well-developed sampling strategy and large survey teams and the willingness of survey respondents to participate and to remind voting decisions 4 or 5 years earlier. As a result, sampling and measurement error and biases loom large—without clearly being exactly known (Schoen 2003; van Elsas et al. 2013).

A second strategy as a low cost alternative is meanwhile often used by offices of statistics. It builds on aggregate data and uses ecological inference (see e.g. Achen and Shively 1995; King 1997). Ecological inference comprises a variety of approaches where the researcher tries to make inferences from an aggregate relationship to a relationship at a lower level, as a rule for the individual level. As aggregated election data is in many cases easily available, this kind of analysis can be performed.

The problem of investigating voter transitions for R parties in an election in time t and C parties in an election in time $t + 1$ can be represented by an $R \times C$ -table (see Table 1). The objective is to estimate correctly the inner cells from the marginals. Those voting in both elections for the same party are called the stayers or loyals, those voting for different parties are called movers or switchers. In our example, the stayers are represented by the cells on the main diagonal, while the cells for the movers are the non-diagonal cells. The central idea of applying ecological inference is the usage of the marginals of the $R \times C$ tables from many electoral districts. From the variation between the districts, estimates for the individual voting behaviour are deduced. Obviously, for multi-party systems, a major barrier to get reliable estimates is the high number of parameters to be estimated.

Furthermore, ecological inference suffers from crucial methodological problems and is notoriously plagued by an indeterminacy problem (Elff et al. 2008; Cho and Manski 2008). In mathematical statistics it is characterized as an ill posed problem, while econometricians consider it as problem with unidentified parameters. One aspect has become famous under the label “ecological fallacy” (Robinson 1950), which captures the fact that relationships observed at the aggregate level possibly do not hold at the individual level. Subsequently, a series of different approaches have been developed for ecological inference. Some models are designed for 2×2 situations and therefore are not directly suitable for estimation of voter transitions in multi-party systems. All proposed approaches fundamentally hinge on their underlying assumptions (Cho 1998; Cho and Manski 2008). Despite more recent overviews (King 1997; King et al. 2004; Wakefield 2004; Cho and Manski 2008) there is no consensus regarding the choice of the methods, nor do we have systematic evidence-based assessments of the statistical performance of these approaches. More specifically, for voter transitions there are ecological inference based studies by Thomsen (1987), Ambühl (2003), Andreadis and Chadjipadelis (2009), Kellermann (2011). In this paper, we present an assessment of ecological inference methods with a special focus on the estimation of

voter transitions. We consider methods which have been applied by practitioners and in science more recently.

The paper is organized as follows: In Sect. 2, we present an overview of the respective statistical approaches including a brief outline of the models and the corresponding assumptions. In Sect. 3, we apply the methods to election data of the German Metropolitan City of Munich to analyse voter transitions between the Federal elections in 2005 and 2009. This is followed by a simulation study to assess the relative performance of the methods detailed above. As practical experience shows that data preprocessing decisions influence the results, the second part of our simulation study quantifies the respective outcomes (Sect. 4). Section 5 gives a discussion of the implication of our results and suggests directions for further research.

2 Ecological inference models for voter transition

We use as an example a $R \times C$ -voter-transition table with R rows and C columns for one district and two elections (see Table 1). The margins, represented by “XX”, are the official results for this district. The unknown values of interest are the inner cells of the table, which are marked with a “?”. [Duncan and Davis \(1953\)](#) emphasised the fact that the margins involve restrictions to the inner cells and as a consequence determine a range of possible values. However, the range of feasible values resulting from these restrictions (“bounds”) is usually wide and therefore often almost non-informative.

The relationships between the two elections can be described by a system of C equations, where C is the number of parties in the second election. E.g., the equation for the CSU₂ in district i has the following form:

$$\text{CSU}_{2,i} = \beta_{\text{CSU,CSU},i} * \text{CSU}_{1,i} + \beta_{\text{SPD,CSU},i} * \text{SPD}_{1,i} + \dots + \beta_{\text{NonVoter,CSU},i} * \text{NonVoters}_{1,i} \quad (1)$$

CSU_{2,i} is the result in district i in the second election, CSU_{1,i} is the corresponding value for the first election. SPD_{1,i} represents the SPD voters and NonVoters_{1,i} denotes the non-voters in the first election in this district. The coefficients $\beta_{\text{party1,party2},i}$ indicate the fraction of voters, which voted in the first election for party1 and in the second election for party2, e.g. $\beta_{\text{SPD,CSU},i}$ stands for the fraction of first election SPD voters, which voted in the second election for the CSU. These equations represent the relationship between both elections ([King 1997](#)). Instead of the counts, e.g. CSU_{2,i}, it is also possible to use the relative frequencies, e.g. $P(\text{CSU}_{2,i})$. Provided the inner cells

Table 1 Available aggregated data in one district

party	CSU ₂	SPD ₂	...	NonVoter ₂	Sum ₂
CSU ₁	?	?	?	?	XX
SPD ₁	?	?	?	?	XX
...	?	?	?	?	XX
NonVoter ₁	?	?	?	?	XX
Sum ₁	XX	XX	XX	XX	XX

of Table 1 consist of fractions of the row margins, they are identical to the β – values in (1).

The main idea of ecological inference is using the data from multiple districts and the reliance on appropriate assumptions. Most assumptions relate to the β –coefficients in (1) and the presumption of some kind of similarity between the districts. While in the classic “Goodman (1953) regression” constant coefficients are assumed, more recent contributions propose complex distributional assumptions on the β –coefficients. Other proposals suggest multinomial logit-related forms (e.g. Thomsen 1987) or provide algorithms gradually adjusting 2×2 extracts of an $R \times C$ table (e.g. Andreadis and Chadjipadelis 2009). Obviously, the appropriateness and flexibility of the model’s underlying assumptions affects the performance of the models.

2.1 Ecological regression

Goodman (1953) introduced regression analysis for the estimation of voter transitions. He assumed constant coefficients across all districts, instead of district specific coefficients (“constancy assumption”).

$$P(\text{CSU}_{2,i}) = \beta_{\text{CSU,CSU}} * P(\text{CSU}_{1,i}) \\ + \beta_{\text{SPD,CSU}} * P(\text{SPD}_{1,i}) + \dots + \beta_{\text{NonVoter,CSU}} * P(\text{NonVoters}_{1,i}) \quad (2)$$

In the $R \times C$ -case, one regression for every party has to be calculated. The coefficients are estimated by least squares using the data of all districts. If the model assumption holds, the coefficients can be directly interpreted as transition rates between the parties. In this case, the coefficients have to be in the range $[0, 1]$. One problem in practical applications is that not all estimated transition rates are guaranteed to be within the admissible range (King 1997), therefore a second step in the estimation process is required to obtain face value valid estimates. We evaluate two different approaches for such an adjustment:

$$\beta_{i,j}^* := \max(0, \min(1, \beta_{i,j})) \quad (3)$$

$$\beta_{i,j}^* := \frac{\beta_{i,j} - \min(0, \beta_{i,1}, \dots, \beta_{i,C})}{\max(1, \beta_{i,1}, \dots, \beta_{i,C}) - \min(0, \beta_{i,1}, \dots, \beta_{i,C})} \quad (4)$$

The first option (3) has been already used by Ambühl (2003). The second alternative (4) is our proposal. Note that in (3) only values outside $[0, 1]$, while in (4) all values are affected. The main advantage of (4) is the preservation of the relative distance between the estimated values. An additional second adjustment is often necessary to fulfil the row restrictions:

$$\beta_{i,j}^{**} := \frac{\beta_{i,j}^*}{\sum_{j=1}^C \beta_{i,j}^*} \quad (5)$$

It is not ensured that the column restrictions are met after (5). Therefore, a final step is necessary to obtain valid tables, which comply with row and column restrictions.

Like [Ambühl \(2003\)](#), we use an iterative proportional fitting algorithm ([Deming and Stephan 1940](#)) for this adjustment. Iterative proportional fitting adjusts alternating the rows and columns of a table until the convergence criterion is met. Mostly, Goodman regression coefficients outside the admissible range indicate a violation of the underlying constancy assumption ([King 1997](#)). The presented adjustments guarantee face value valid final tables. However, their statistical properties remain unclear.

For a long time, the Goodman model and its extensions have been considered the standard and are still used in practice, e.g. for the recent estimation of voter transitions in the 2013 Austrian national elections ([SORA 2013](#)). As candidate model for our assessment, we use the basic ecological regression model presented in (2). Because our districts are not equally sized, one could argue that the district size should be used as weights. However, this method is not undisputed ([King 1997](#)). As a result, both weighted and unweighted regressions are considered as potential models for our assessment.

2.2 Hierarchical models

Starting with the book by [King \(1997\)](#), a new type of model for ecological inference was introduced. For the 2×2 -case, the author proposes a regression based approach using the relationship (1) and the parameter restrictions induced by the margins. He estimates the transition rates using two different levels, the district-connecting level and the district level. At the second level, connecting the districts, a similarity between all districts is assumed. This similarity is expressed by a common truncated bivariate normal distribution of the district transition rates. For the first level, the district level, he employs this truncated bivariate normal distribution and the information of the margins of the district to estimate the district level transition rates ([King 1997](#)). Then, hierarchical Bayesian models for the 2×2 -case ([King et al. 1999](#)) and for the $R \times C$ -case ([Rosen et al. 2001](#)) have been proposed.

Hierarchical Bayesian ecological inference models share the basic principle of the proposal in King's book, that the estimation of voter transition is separated in different levels. The upper level represents the district level, while the lower level combines the information over all districts assuming a specific distribution for all district transition rates. For the parameter estimation, Bayesian Markov Chain Monte Carlo (MCMC) principles are applied. One proposed $R \times C$ -model is called "hierarchical multinomial-Dirichlet model". Here, at the district level, a multinomial distribution is assumed ([Rosen et al. 2001](#)).

$$\begin{aligned} &(\text{CSU}_{2,i}, \text{SPD}_{2,i}, \dots, \text{NonVoters}_{2,i}) \\ &\sim \text{Multinomial}(\theta_{\text{CSU},i}, \theta_{\text{SPD},i}, \dots, \theta_{\text{NonVoters},i}, N_i) \end{aligned} \quad (6)$$

The θ s are assumed to be weighted averages of the individual transition rates, e.g. for the CSU in district i :

$$\theta_{\text{CSU},i} = \frac{\beta_{\text{CSU},\text{CSU},i} * \text{CSU}_{1,i} + \dots + \beta_{\text{NonVoter},\text{CSU},i} * \text{NonVoters}_{1,i}}{\text{Voters}_{1,i}} \quad (7)$$

At the second level, a common Dirichlet distribution of the β s over all districts is assumed for each row:

$$(\beta_{\text{CSU,CSU},i}, \beta_{\text{CSU,SPD},i}, \dots, \beta_{\text{CSU,NonVoter},i}) \\ \sim \text{Dirichlet}(\alpha_{\text{CSU,CSU}}, \alpha_{\text{CSU,SPD}}, \dots, \alpha_{\text{CSU,NonVoter}}) \quad (8)$$

Rosen et al. (2001) propose as hyperpriors for the α s an exponential distribution. Lau et al. (2007) implement this model however employing a gamma distribution in their R-library (Lau et al. 2012) for R Core Team (2013). We use this implementation in our assessment.

The hierarchical multinomial-Dirichlet model is only one potential solution in the class of hierarchical models. Different distributional assumptions for the different levels have been proposed. One example is the hierarchical multinomial-normal model proposed by Greiner and Quinn (2009) who also provide an R-package for their model. The distributional assumptions for this model are presented in the appendix. Both models apply MCMC methods and require the appropriate choice of sample size, burnin and thinning. Inappropriate settings for these values, especially for the burnin, often lead to not converging chains and as a result to dubious estimates.

2.3 Thomsen's probit/logit approach

Thomsen (1987) proposes a quite distinct approach based on latent variables. For the 2×2 -case, i.e. the choice between two parties, the tetrachoric correlation ρ_T is offered for the estimation of the inner cells of the table. Using two parties, P_1 and P_2 , for the first and second elections the following relationship should hold

$$P(P_{11}, P_{12}) = \Phi_2(\Phi^{-1}(P_{11}), \Phi^{-1}(P_{12}), \rho_T). \quad (9)$$

Here Φ and Φ_2 denote the standard normal and the two-dimensional standard normal with correlation ρ_T , respectively. $P(P_{11}, P_{12})$ is one inner cell of the table and all other inner cells can be directly calculated from $P(P_{11}, P_{12})$ and the margins $P_{11}, P_{12}, P_{21} = 1 - P_{11}, P_{22} = 1 - P_{12}$.

The key assumption of Thomsen is that the ecological probit correlation ρ_e between districts can be used as an estimator for the overall tetrachoric correlation ρ_T .

$$\hat{\rho}_T = \rho_e = \text{corr}(\Phi^{-1}(P_{11}^*), \Phi^{-1}(P_{12}^*)) \quad (10)$$

P_{11}^* and P_{12}^* in (10) denote the marginal party share vectors for the districts.

This model cannot easily be generalized for $R \times C$ -tables, since (10) does not longer hold for 2×2 -tables, which result from a collapsed $R \times C$ -table (Thomsen 1987). Therefore, the author proposes a complex algorithm using his model for 2×2 -subtables with an estimation of the corresponding (unknown) margins. In this algorithm, it is necessary to define a reference party for each election. Thomsen proposes the use of the non-voters as a neutral category, but there is no theoretical reasoning behind this proposal (Thomsen 1987).

Thomsen et al. (1995) provide a software solution, which offers a second estimation method. His “complete algorithm” represents an advancement of the original “incomplete algorithm” and equates the tetrachoric correlation of each district individually with the ecological correlation over the districts. It allows district specific estimates. A second addition in the software is the possibility of an adjustment for sampling processes.

This model has been criticized by Ambühl (2003) and discussed and expanded by Park (2008). Due to the availability of the software solution and the relative short estimation times, Thomsen ecological inference approach has been frequently used by the offices of statistics of German cities for the calculation of voter transitions.

2.4 Recursive algorithm for 2×2 -approaches

Another approach for the estimation of voter transitions has been proposed by Andreadis and Chadjipadelis (2009). Their recursive algorithm expands 2×2 -ecological inference models to the general $R \times C$ -case by a consecutive consideration of collapsed 2×2 -tables. The choice of the inner cell (i, j) for collapsing the table, which is called the “best pair”, is performed by:

$$\begin{aligned} \text{best pair} &= \underset{(i,j)}{\operatorname{argmax}}(C_{i,j}^*) \\ \text{with } C^* &= (P(\text{CSU}_1), P(\text{SPD}_1), \dots, P(\text{NonVoter}_1)) \\ &\quad \times \operatorname{corr}(E_1, E_2) \times (P(\text{CSU}_2), P(\text{SPD}_2), \dots, P(\text{NonVoter}_2))^T \quad (11) \end{aligned}$$

$\operatorname{corr}(E_1, E_2)$ denotes the matrix of correlation of the observed election results E_1 and E_2 in the districts between the two elections.

Then the value $h_{i,j}$ in cell (i, j) is estimated by the corresponding 2×2 method from the collapsed table. The algorithm is iterated by subtracting $h_{i,j}$ from the marginal distribution and the selection of a new “best pair” using (11) and the new margins. The algorithm continues until the voter transition of a sufficient percentage of voters is estimated. The authors claim that one potential advantage of the algorithm is an improved performance when estimating small transitions. In the underlying transition table, the margins are suggested to rarely impose restrictions on small voter transition. However, as each iteration of their algorithm reduces the number of voters, whose voter transitions have to be estimated, the margins in later iterations may impose restrictions on small voter transitions (Andreadis and Chadjipadelis 2009).

For the estimation of the selected 2×2 -tables, the authors rely on a model proposed by Grofman and Merrill (2004). Grofman and Merrill’s 2×2 -model utilizes a special relationship in the 2×2 -case. All possible pairs $(\beta_{\text{CSU}_1, \text{CSU}_2}, \beta_{\overline{\text{CSU}_1}, \text{CSU}_2})$ of the collapsed table $(\text{CSU}_1, \overline{\text{CSU}_1}) \times (\text{CSU}_2, \overline{\text{CSU}_2})$ must lie on one straight line. This line is called “tomography line” (King 1997) and is defined by the marginal election results $P(\text{CSU}_1), P(\overline{\text{CSU}_1})$ and $P(\text{CSU}_2)$.

$$\beta_{\overline{\text{CSU}_1}, \text{CSU}_2} = -\frac{P(\text{CSU}_1)}{P(\overline{\text{CSU}_1})} \beta_{\text{CSU}_1, \text{CSU}_2} + \frac{P(\text{CSU}_2)}{P(\overline{\text{CSU}_1})} \quad (12)$$

A tomography line exists for every district, but can also be calculated for the whole geographic area. The suggested model minimizes the quadratic distance between all district tomography lines and the whole area tomography line (Grofman and Merrill 2004).

Andreadis and Chadjipadelis claim that for the “task in our method any of the known methods for the solution of the 2×2 ecological inference problem could be used” (Andreadis and Chadjipadelis 2009). The authors justify their choice with the argument that the Grofman and Merrill model has similar results to the King model from 1997, but that the former is easier to implement and has lower computational complexity. Both models are based on the restrictions imposed by the margins, therefore it shall be “better to spend available resources to increase the number of statistical (electoral) units instead of spending resources on complicated, sophisticated and intensive methods to model data” (Andreadis and Chadjipadelis 2009).

A modification for the algorithm has been proposed by Kellermann (2011). He alters the order of selected best pairs in the first few iterations of the algorithm to enforce that at first each loyalty rate is considered once. He assumes that loyalty rates tend to be higher and that the forced estimation of all loyalty rates enhances the algorithm. This version has been used by several offices of statistics in Germany.

3 A comparison of ecological inference approaches

Our evaluation includes the six discussed models (see Table 2) and is conducted in two consecutive steps. In the first step, we compare the results from the different models for the federal election 2005 (Statistisches Amt der Landeshauptstadt München 2005) and 2009 (Statistisches Amt der Landeshauptstadt München 2009) in Munich. This gives us useful insights on the practical problems using the different approaches. In the second step, we proceed with an extensive simulation study comparing the performance of the different estimation procedures.

To compare different voter transitions tables, we use the absolute distance (AD) between the voter transition tables in shares. More specifically, we calculate the sum of

Table 2 Selected models for the assessment

Short name	Model name	Sections
Goodman	Goodman’s ecological regression	Section 2.1
eiPack	Rosen et al.’s hierarchical multinomial-Dirichlet model in Lau et al.’s implementation for R	Section 2.2
G–Q	Greiner and Quinn’s hierarchical multinomial-normal model not included in simulation, see Section 3.1	Section 2.2
Thomsen	Thomsen’s logit approach	Section 2.3
And and Cha	Andreadis and Chadjipadelis’s recursive algorithm with Grofman and Merrill’s 2×2 -model	Section 2.4
Kellermann	Kellermann’s adjusted recursive algorithm with Grofman and Merrill’s 2×2 -model	Section 2.4

Short name (used in graphics), long model name and section with model description

all absolute differences of all values of the transition tables. Given two voter transition tables A_1 and A_2 , each with R rows and C columns, the $AD(A_1, A_2)$ is defined by:

$$AD(A_1, A_2) = \sum_{r=1}^R \sum_{c=1}^C |A_1(r, c) - A_2(r, c)| \quad (13)$$

The AD has the range 0–2 and can be interpreted as follows: half of the value represents the fraction of voters, which must be shifted within one table in order to achieve two identical tables. A similar measurement has already been employed by [Thomsen \(1987\)](#).

3.1 Model evaluations for Munich election data

We analyze the Munich election data from the federal elections 2005 and 2009 at the district level. We consider seven possible outcomes (“parties”) in both elections: CSU, SPD, FPD, GRÜNE, “DieLinke”, “Others” and non-voters. “Others” is the cumulated category of the remaining small parties. For the analysis we only use the second vote (party vote, German: Zweitstimme). Voters by mail are added to their corresponding ballot box districts. Due to changes in the boundaries of the districts, it is necessary to form time constant regions. This additional aggregation leads to a reduced number of available districts for the analysis. In total, 314 time constant districts remain for the analysis. They were created from 654 districts in 2005 and 664 districts in 2009. Most time constant districts are of similar size (median: 1376.5 eligible voters), with several exceptions (up to: 56,926 eligible voters). A short description of the two elections is provided in the appendix. Since all our models include different possible settings and parameters, we compare the results of the same model with different settings. This within-model comparison gives an indication for the stability of the models.

For the Goodman regression, we compare weighted and unweighted regression. Furthermore, we consider the two presented out-of-admissible-range corrections of coefficients. We do not find large differences between weighted and unweighted regression. Coefficients outside the admissible range are observed after the first estimation step. Therefore, a correction of the estimated parameters is necessary (see Sect. 2.1). We observe relevant differences in the estimated voter transitions depending on the used out-of-admissible-range correction (see Table 9 in the appendix for details). Since approach (3) results in many structural zeros, we consider (4) as more suitable. Therefore, we rely on the unweighted regression and the adjustment of estimates is performed using the approach (4).

We include two hierarchical models: the hierarchical multinomial-Dirichlet model implemented in the R-library “eiPack” and Greiner and Quinn’s hierarchical multinomial-normal model implemented in their R-library “ $R \times C$ EcolInf”. Using Bayesian hierarchical models with MCMC algorithms requires the selection of appropriate values for burnin, thinning and chain length. For both included hierarchical models, the default values for burnin and thinning are inadequate for our data. However, quite stable chains are achieved for the hierarchical multinomial-Dirichlet model (eiPack) using an adequate burnin and chain length. Due to a strong autocorrelation

in the chains, especially for cells with small expected values, only very high thinning values reduce the in-chain autocorrelation. While, in general, the models have a good convergence, in some chains of the smaller values a local deflection from the global chain mean may be observed. In case of the Greiner and Quinn model, we are not able to select suitable values for burnin and sample size. We also do not achieve convergence within an acceptable computation time. Therefore, the Greiner and Quinn model is excluded from all further examinations. As values for the hierarchical multinomial-Dirichlet model (eiPack), we find the usage of standard priors, a burnin of 50,000, a sample size of 1000 and a thinning of 1000 suitable. This setting gives us good results in the trade-off between calculation time and estimation quality. In comparison to longer chains, we observe only minor differences between the results (see the appendix for details).

A requirement of Thomsen's model is to indicate a reference party for each election, see Sect. 2.3. This selection has a major impact on the obtained estimates (see the appendix for details). One additional important parameter has been the selected estimation method. We observe relevant differences between the results of the incomplete and the complete algorithm ($AD = 0.30$). Furthermore, the numbers of iterations and adjustment for sample processes in smaller districts have a reduced impact on the result. For the following, we select the non-voters as reference party for both elections, the incomplete algorithm and adjustment of sample processes.

The recursive algorithm proposed by Andreadis and Chadjipadelis (And and Cha) uses the respective 2×2 -model as a "parameter" to be chosen. In addition to the Grofman and Merrill model implemented by Andreadis and Chadjipadelis (see Sect. 2.4), we also apply the hierarchical multinomial-Dirichlet model (eiPack) model as an alternative 2×2 -model (see Sect. 2.2). We find large differences between the estimates using the different 2×2 -models ($AD = 0.32$). A close inspection reveals that the selected "best pairs" in the first few iterations are different. We also observe some implausibly high loyalty estimates using both 2×2 -models where values up to 0.99 are estimated. In summary, the approach shows a high instability depending on the settings used. To be as close as possible to the original proposed algorithm and also the practical application of this model, we use as 2×2 -model for the algorithm the Grofman and Merrill model implemented by Andreadis and Chadjipadelis.

In the next step, we perform a comparison between the different models. Table 3 provides the distances between the model estimates for the federal elections 2005

Table 3 Comparison of the estimated voter transitions between the models, Federal election 2005 and 2009 in Munich, AD

	Goodman	eiPack	Thomsen	And and Cha	Kellermann
Goodman	0.00	0.54	0.36	0.51	0.52
eiPack	0.54	0.00	0.34	0.25	0.25
Thomsen	0.36	0.34	0.00	0.36	0.37
And and Cha	0.51	0.25	0.36	0.00	0.10
Kellermann	0.52	0.25	0.37	0.10	0.00

Estimated Voter Transition for the Federal Election 2005 and 2009 in Munich

2005	CSU	0.756	0.013	0.017	0.01	0.007	0.011	0.185
	SPD	0.017	0.562	0.023	0.011	0.069	0.141	0.177
	FDP	0.021	0.02	0.883	0.019	0.017	0.019	0.02
	GRÜNE	0.02	0.019	0.037	0.877	0.012	0.015	0.019
	DieLinke	0.033	0.036	0.035	0.037	0.731	0.078	0.05
	Others	0.058	0.064	0.046	0.039	0.221	0.425	0.147
	NonVoters	0.075	0.05	0.121	0.095	0.018	0.013	0.628
		CSU	SPD	FDP	GRÜNE	DieLinke	Others	NonVoters
		2009						

Fig. 1 Estimated voter transition for the Federal election 2005 and 2009 in Munich, based on hierarchical multinomial-Dirichlet Model Implemented in eiPack

and 2009. The smallest AD can be found between both recursive models with an AD of 0.10. Only five percent of all voters are differently allocated. The estimates from the hierarchical multinomial-Dirichlet model are relatively close to these results, but the AD of 0.25 is considerably higher. Thomsen's model seems to have a similar distance to all other models, but with an AD of 0.34 the estimate is already more distinct. The estimates from the Goodman model are the most distinct as most ADs are around 0.5.

Figure 1 presents the estimated voter transition based on the hierarchical multinomial-Dirichlet model implemented in eiPack. The values constitute the transition rates. Each row adds up to one and describes the transition of the 2005 voters. For example, the model estimates a loyalty rate of 75.6 % for the CSU, whilst 18.5 % of the 2005 CSU voters were non-voters in the 2009 election.

Our results clearly indicate that some of the settings of the different model have a high impact on the estimation of the voter transitions. They lead to remarkable differences within the model when applied to identical data. A second finding is that there are also differences between the models, which can be narrow such as between the two recursive algorithms, or major such as between Goodman and all other models. These remarkable differences raise the question of which model actually is the best one. This will be tested using a simulation in the following section.

3.2 Simulation-based assessment of model performance

The following subsection presents the main assessment of our model comparison. In order for the study to have a realistic design, we use the original Munich election results from the elections in 2005 and 2009 for 314 time constant districts. The assumed voter transitions are derived from a federal voter transition table published by Infratest dimap. The results of the second election are then simulated at the individual level using different voter transitions structures. The margins from the district voter transitions tables are used as input data for the ecological inference models. The simulated individual level voter transitions table at the town level is used as the reference and is compared to the estimated voter transition tables resulting from each of the respective methods. Our simulation design includes four different scenarios, which reflect different assumptions for the distribution of the voter transitions on the district level.

In the first scenario (I.1), we use constant transition rates over all districts. The district-specific results are multinomially distributed with the global transition rates. The second scenario (I.2) removes the constancy of the transition rates over the districts. The district-specific transition rates follow a common Dirichlet-distribution. The third scenario (I.3) expands the second scenario (I.2). Instead of one common Dirichlet distribution, there are three different population groups in each district with a group-specific common Dirichlet-distribution over the districts. The respective groups differ mainly in their party loyalty. The last scenario (I.4) again expands the second scenario (I.2) and addresses the problem of aggregation bias. Instead of a common Dirichlet-distribution, the mean of the district-specific Dirichlet-distribution depends on the success of a party in the first election. We lower the loyalty of SPD-voters depending on the combined success of the SPD and “DieLinke” in this scenario. The CSU exhibits an increased loyalty depending on the combined success of CSU and FDP. Furthermore, we apply an admissible range for the calculated loyalty rates to avoid implausible values. Table 4 presents a summary of these four scenarios. Note, that scenario I.1 and I.2 are created following the distributional assumption of the Goodman regression and the hierarchical multinomial-Dirichlet model, respectively. However, scenarios I.3 and I.4 do not comply with the distributional assumptions of any model. In scenario I.3, a mixture of three Dirichlet-distributions is assumed for each district and we suppose district-specific Dirichlet-distribution for two rows in scenario I.4.

Table 4 Summary of key features of the data scenarios for model assessment

Scenario	Name	Distribution within district	Distribution of district parameters	Different populations	Aggregations bias
I.1	Goodman	Multinomial	Non (constant)	No	No
I.2	eiPack	Multinomial	Dirichlet	No	No
I.3	Populations	Multinomial	Dirichlet (mixture)	Yes (3)	No
I.4	Aggregation Bias	Multinomial	Dirichlet (specific)	No	Yes

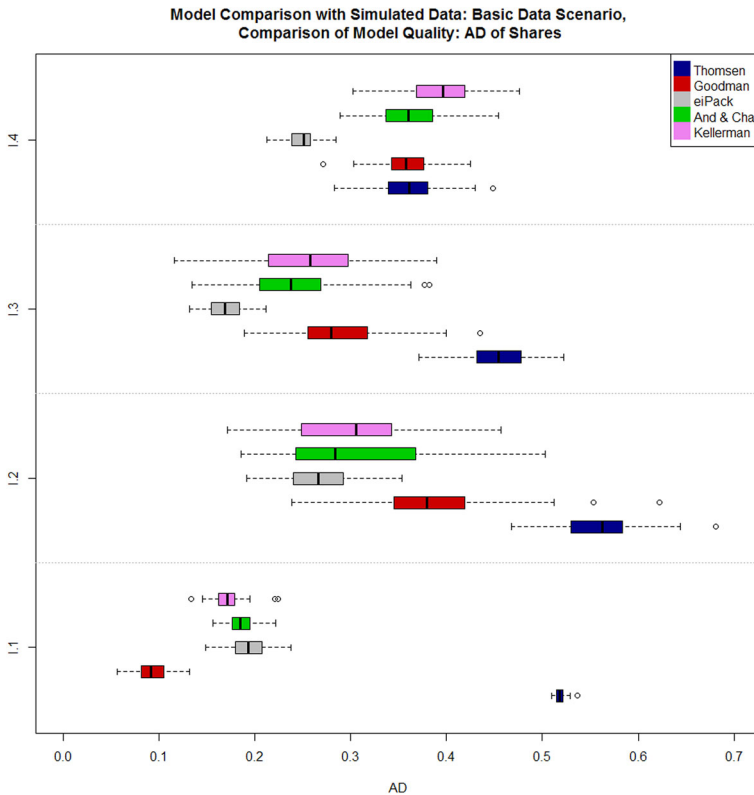


Fig. 2 Model comparison based on simulation study: basic scenario, comparison of model quality: AD of shares, 100 simulation for each scenario

For each scenario, we perform 100 repetitions and estimate voter transitions using the five selected methods. We use the AD between the true values calculated from the underlying simulated individual level data and the estimated voter transition at the town level as our assessment criteria. In Fig. 2, we present the boxplots of the ADs for the five models and four scenarios.

In the first scenario (I.1), low ADs occur (see Fig. 2) and the Goodman model has the lowest AD. This is no surprise as the Goodman assumption is fulfilled. The hierarchical model and both recursive approaches (And and Cha, and Kellermann) have estimates of comparable quality. The Thomsen model shows the worst performance of all models. In the second scenario (I.2), the overall performance decreases. The hierarchical model has the lowest ADs, but even there are on average 14 % of all voters wrongly allocated. A somewhat less favorable result can be found for both recursive models. The Goodman model clearly negatively distinguishes itself from these three models. Again, the Thomsen model exhibits the overall worst performance in this scenario. In comparison to the second scenario, the ADs are lower in the third scenario (I.3). The lowest ADs in this scenario has the hierarchical model, followed by the recursive approaches and the Goodman model. Again, the Thomsen model,

while showing lower ADs than in the first two scenarios, excels negatively. In the fourth scenario (I.4), the estimates for all models show high ADs. The hierarchical model clearly outperforms all other models and has a performance comparable with the second scenario. The performance of the other four models is similar in this scenario.

Summarizing, the Thomsen model has the worst performance of all models in our comparison. The good results of the Goodman model in scenario I.1 and the worse results in the other scenarios prove that iff the assumption of the model holds then the models' estimates are usable. The recursive models have a similar performance. However, whilst they are superior to the Thomsen and Goodman overall, the hierarchical multinomial-Dirichlet model clearly outperforms them. This is especially pronounced in scenarios I.3 and I.4, which can be considered the most realistic and complex scenarios in this comparison. Additional examinations using other performance measurements, for example the absolute distance calculated based on the voter transition rates, were also carried out. In general, the results are quite similar in structure to the presented assessment.

4 Data preprocessing issues

Surprisingly, most studies dealing with ecological inference approaches do not bring up the subtleties of data preprocessing strategies. However, real-world applications require a series of decisions relating to this, which may have enormous consequences as we will demonstrate below. While we only examine typical voter transition issues, we believe that the insights gained in this analysis are not limited to this specific case.

4.1 Violation of population assumptions in voter transition applications

One key assumption for applying ecological inference using data from electoral districts is that the voters do not change between the elections. This assumption is potentially invalid for two reasons. Firstly, there is a change of the population in the electoral districts and secondly voters can decide to vote by mail.

4.1.1 Change of the delineation of administrative boundaries and of the composition of the population

Between two elections the layout of electoral districts may change. Some districts can be merged and some can be split. Even a complete rearrangement of subregions is possible. The magnitude of such changes differs between regions and the elections in question. However, aggregation of electoral districts to time constant areas is almost always necessary. Of course, such steps introduce additional sources of error and reduce the number of districts available for the analysis.

Furthermore, there are also changes in the electorate. A voter can be entitled to vote for the first time in the second election or die between the elections. If a person is eligible to vote in both elections, he or she can change his electoral district by moving to another location between two elections. It is also possible that migration leads to additional changes in the population size of the districts. It can be concluded that not

Table 5 Available aggregated data in one district, expanded with changes in population

party	CSU ₂	SPD ₂	...	NonVoter ₂	Died	movingOut	Sum ₂
CSU ₁	?	?	?	?	?	?	XX
SPD ₁	?	?	?	?	?	?	XX
...	?	?	?	?	?	?	XX
NonVoter ₁	?	?	?	?	?	?	XX
newVoters	?	?	?	?	0	0	XX
movingIn	?	?	?	?	0	0	XX
Sum ₁	XX	XX	XX	XX	XX	XX	XX

every voter will be an eligible voter in both considered elections. For example, ca. 100,000 residents moved within Munich and around the same number left the town during 2013. In the same period, approximately ca. 125,000 moved into the town in the same period ([Statistisches Amt der Landeshauptstadt München 2014](#)). In total, these movers correspond to around 15 % of the population.

Due to the ecological inference models' use of both election results of the district to estimate voter transition, such changes in the electorate of the districts represent an additional source of errors. Arguing that voters leaving an electoral district are replaced by voters with similar opinions, would allow to ignore population changes.

Table 5 illustrates voter transitions in one district in a more precise manner. It is an expanded version of Table 1 with additional rows and columns for the population changes in the electoral districts described above. Some of the inner cells are structural nulls, e.g. a first time voter in the second election cannot be a voter of the first election who died in the meanwhile. Comparing Tables 1 and 5, it is obvious that there are differences in the numbers. For example, the margins of Table 5 are, in general, not identical to the margins of Table 1.

4.1.2 Vote by mail

Another major difficulty arises from voters by mail. The electoral region is divided into electoral districts (ballot box districts) and each voter is affiliated with one of this districts. Voters by mail are also part of these districts, but their results are presented in special postal vote districts. Postal vote districts are, in general, not identical with one ballot box district. It is more common that two or three ballot box districts form one postal vote district. In some cases, only a considerably less precise assignment is possible. In addition, being a voter by mail is not necessarily a constant feature of a voter: In one election a person can vote by mail, in the next election the same person can vote in the respective ballot box district. Furthermore, there is a major difference in the election participation. The turnout of people requesting voting by mail is much higher than the ballot box turnout. A person who actively requests voting by mail is more likely to participate in an election. Moreover, it is common for the elections results of voters by mail and ballot box voters to be different. In Germany, voting by

mail is not a negligible problem, as over 40 % of all voters voted by mail in Munich in the federal election 2013 ([Statistisches Amt der Landeshauptstadt München 2013](#)).

4.2 Evaluation of different data preprocessing strategies

To deal with the problems outlined in Sect. 4.1, data preprocessing is necessary. In total, we assess ten different data preprocessing strategies which are all based on the same true aggregated election results for the Munich federal election 2005 and 2009 at the district level. We consider four different data preprocessing strategies covering the issues discussed in Sect. 4.1 and six additional data preprocessing options to address additional questions. As mentioned in Sect. 3.1, 314 time constant districts are available and we consider seven parties in each election.

One question resulting from the data is the handling of population changes (see Sect. 4.1.1) in voting districts. We apply two approaches: We use the maximum of eligible voters over both elections (a). The difference between the maximum and the actual observed eligible voters is added to the non-voters, while the number of voters for each party stays unadjusted. The non-voters in this case also cover the net total of all population changes. Our second approach (b) has been proposed by Hawkes ([Kellermann 2011](#)). In this approach, the number of voters for the parties is adjusted, keeping the observed shares.

A second aspect concerns the handling of voters by mail (see Sect. 4.1.2). We use two different approaches, our first approach (1) ignores the voters by mail completely, while the second approach (2) distributes the voters by mail to their ballot box districts proportional to the number of eligible voters in this districts. For the 2005 election, it was only possible to distribute the voters by mail within the municipality. For the 2009 election, a postal vote district corresponds to two or three ballot box districts.

Both data preprocessing steps can be combined and as a result we have four different datasets. The data already used in Sect. 3.1 is scenarios (2a): distributing the voters by mail (2) and absorbing population changes into the non-voters (a). We call this dataset our base preprocessing dataset. In Table 6, the differences between the estimated voter transitions for the Munich election data are presented. As a measurement for the distance, we again use the AD (see Sect. 3). The results for the Kellermann adjustment of Andreadis and Chadjipadelis (And and Cha) algorithm are not presented, because they are almost identical to the values of the original algorithm. Note that all estimated voter transition shall describe the same election, only the data preprocessing is different.

It is obvious that the estimated voter transitions are not identical as ADs up to 0.42 can be observed. This means that up to 21 % of all voters must be shifted to have identical tables. It seems that models, where only one aspect is changed, are more similar to each other, whilst distances between the models with the highest difference considering the data preprocessing are higher. Some of the observed results can be explained by differing interpretation of the voter transition caused by the different handling of population differences.

Six additional data preprocessing strategies are analyzed using the base preprocessing dataset (2a). The level of aggregation is one question, which arises from the work to find time constant districts, and which also highly influences the calculation time of

Table 6 Comparison of the results between the data preprocessing strategies covering voters by mail and population differences, AD

Goodman	1a	1b	2a	2b	Thomsen	1a	1b	2a	2b
1a	0.00	0.25	0.25	0.33	1a	0.00	0.27	0.18	0.40
1b	0.25	0.00	0.42	0.19	1b	0.27	0.00	0.31	0.21
2a	0.25	0.42	0.00	0.39	2a	0.18	0.31	0.00	0.40
2b	0.33	0.19	0.39	0.00	2b	0.40	0.21	0.40	0.00
eiPack	1a	1b	2a	2b	And and Cha	1a	1b	2a	2b
1a	0.00	0.15	0.20	0.29	1a	0.00	0.22	0.37	0.37
1b	0.15	0.00	0.25	0.19	1b	0.22	0.00	0.42	0.20
2a	0.20	0.25	0.00	0.25	2a	0.37	0.42	0.00	0.36
2b	0.29	0.19	0.25	0.00	2b	0.37	0.20	0.36	0.00

Table 7 Results of the additional data preprocessing strategies, AD between results for the preprocessing strategy and the base preprocessing dataset (2a), preprocessing strategy were applied to the base preprocessing dataset (2a)

Model	2a	3	4	5	6	7	8
Goodman	0.00	0.16	0.34	0.19	0.19	0.22	0.00
eiPack	0.00	0.34	0.75	0.47	0.54	0.14	0.01
Thomsen	0.00	0.06	0.37	0.09	0.14	0.07	0.00
And and Cha	0.00	0.11	0.61	0.19	0.30	0.21	0.00
Kellermann	0.00	0.09	0.15	0.09	0.26	0.21	0.00

some models. In addition to the 314 district model, we also calculate a sixty observation model (3) using an aggregation level proposed by the Munich office of statistics and a municipality model with 25 “districts” (4). The latter can also be considered as model with exact voters by mail distributions. In addition to these models at the town level, we also consider separate models for subregions. Such an assumption does not necessarily hold for the entire region. Therefore, we split the town in five subregions. One split uses the geographical structure of Munich (5) and leads to regions, which can be described as North, South, West, East and Center. A second split uses the election results of the conservative camp. The quantiles of the combined share of CSU and FDP are used to define the regions (6). Furthermore, we change the number of parties and the order of the parties. To test the first, we add the FDP and “DieLinke” to our “Others” party and reduce the 7×7 -problem to a 5×5 -problem (7). To test the influence of the party order, we randomly reorder the party order in one election (8). Therefore, the high loyalty rates are no longer at the diagonal of the voter transition table. The estimated voter transitions are compared with the estimates from the base preprocessing dataset (2a) and the ADs are presented in Table 7.

We find that changing the order of the parties (8) has (almost) no influence on the estimates as both results can be seen as identical. Furthermore the dissimilarity between the results for one model are, in general, at a lower level than the differences between

Table 8 Data scenarios for assessment of data preprocessing

Scenario	Name	Subscenarios	Descriptions
II.1	Aggregation60	4	Additional aggregation to 60 districts
II.2	New party	2	Additional party in election two, 7×8 -table
II.3	Voters by mail	3	Consideration of voters by mail
II.4	Population change	3	Consideration of population changes

the models and the different datasets. One exception is the hierarchical multinomial-Dirichlet model, where most of the results clearly differ from the estimates based on the base preprocessing dataset (2a). Beside strategy (8), only the strategy with a reduced number of parties (7) provides similar estimates. All other strategies have ADs higher than 0.3 up to the maximum of 0.75. A close inspection of the estimated voter transitions reveals that the number of districts has a relevant impact on the estimates. The loyalty rates are estimated to be much lower with fewer observations. This is one of the main reasons for the observed dissimilarity. The subregions models (5 and 6) are also affected by this phenomenon, because each subregion model uses a much lower number of districts. An effect of higher aggregation is visible for almost all considered models. While the reduction to sixty observations (3) still has only a relatively minor impact, the models with municipality level data (4) have, in general, a relatively high distance from the corresponding estimates based on the base preprocessing dataset (2a). The especially high value of 0.61 for Andreadis and Chadjipadelis's algorithm is caused by a different order of selected "best pairs" in the first few iterations of the algorithm.

4.3 Simulation-based assessment of data preprocessing issues

The results presented in Sect. 4.2 prove that data preprocessing matters. The following second assessment covers some of the issues raised. Table 8 presents a short summary of the considered scenarios and a detailed explanation of the scenarios follows in the corresponding subsection. Due to the necessary computational time, only 30 repetitions for each scenario are used. The AD-boxplots for all scenarios are presented in Fig. 3.

4.3.1 Higher aggregation and new parties

The first scenario (II.1) uses data at a higher aggregated level. For this scenario, we use the data from the scenarios I.1–I.4 (see Table 4) and aggregate them to the medium aggregation level already considered in Sect. 4.2. The last number of the subscenarios marks the corresponding scenario in Sect. 3.2. In the second scenario (II.2), we consider the effect of a new party. We simulate new election results with an additional minor party in the second election for the scenarios I.2 and I.4 (see Table 4). Instead of a 7×7 ecological inference problem, a 7×8 ecological inference problem is considered.

The ADs of these simulations presented in Fig. 3 depict that the usage of higher aggregated data (II.1) is leading to a loss of performance. Only the Thomsen model's

**Model Comparison with Simulated Data: Additional Considerations,
Comparison of Model Quality: AD of Shares**

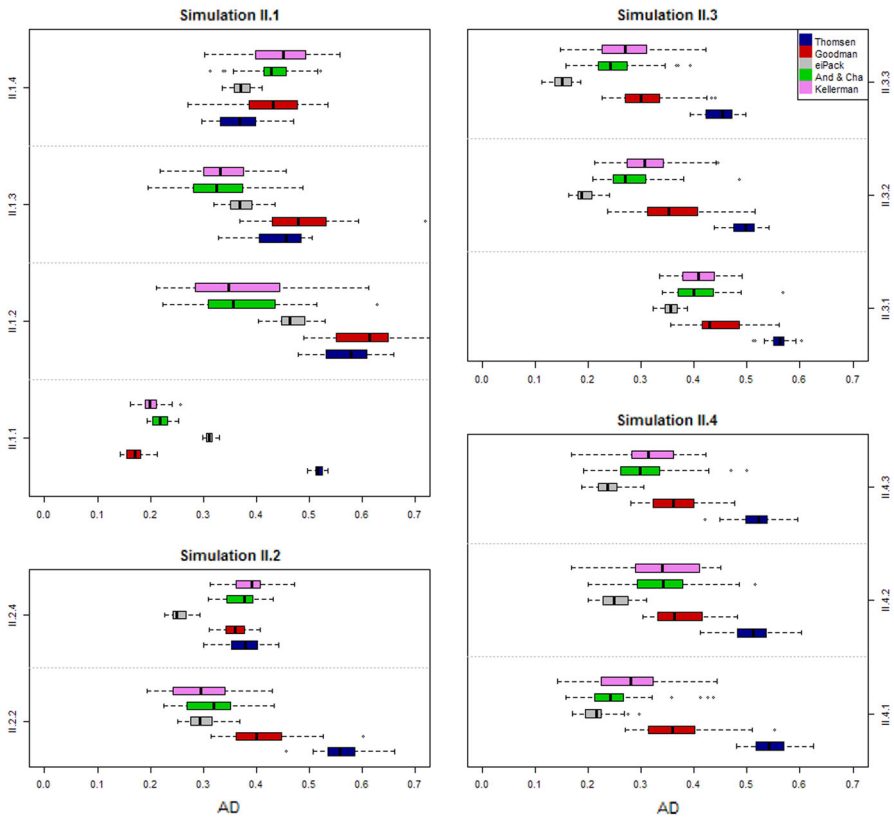


Fig. 3 Model comparison based on simulation study: additional considerations, comparison of model quality: AD of shares, 30 simulation for each scenario

estimates still have a similar quality as all the other models have higher ADs. The strongest negative effect can be identified for the Goodman regression and the hierarchical model. The loss of performance in the simulation ranges from 0.1 to 0.2 AD. The recursive models have a lower performance loss. Due to the loss of performance, the hierarchical model is no longer the dominating model in this scenario. In the second scenario (II.2), most models estimates have a comparable quality. Slightly worse estimates occur for the Goodman regression and the hierarchical model in the scenario II.2.2. Thomsen and Kellermann have on average slightly better estimates in this scenario. However overall, the changes are low and no relevant differences in the performance of the models are visible.

4.3.2 Voting by mail

The third scenario (II.3) is based on scenario I.3 (see Table 4) and adds voters by mail. We assume that on average voters by mail have a higher loyalty, therefore 50 % of the

voters by mail are taken from the high loyalty population in scenario I.3. Furthermore, it is assumed that at average 50 % of voters by mail in the first election are also voters by mail in the second election.

We consider three different scenarios. In the first subscenario (II.3.1), the voters by mail are ignored and the voter transition is estimated using only the ballot box voters. In the second subscenario (II.3.2) we redistribute the voters by mail, similarly to Sect. 3.2, in the corresponding ballot box districts within a municipality. We assume that a much stronger connection is available for the third subscenario (II.3.3). The voters by mail from 3 to 4 ballot box districts form a postal vote district. In addition, the exact number of registered voters by mail in each ballot box district is known and used for the distribution. We reject the idea to use the postal vote districts as additional districts for ecological inference in the simulation. The reasons are described in Sect. 4.1.2.

Ignoring the voters by mail (II.3.1) leads to worse estimates (see Fig. 3) and all models are affected. But while the strongest effect can be found for the hierarchical model, it is—similar to scenario I.3—still the best model in this scenario. Distributing voters by mail at the municipality level (II.3.2) results in slightly worse results compared to scenario I.3. All models are equally affected. If the distribution of the voters by mail can be done at a finer lever (II.3.3), no loss of performance occurs. Therefore, the hierarchical model also has the best performance in this scenario. Furthermore, ignoring voters by mail seems to be the worst option, if the target of the analysis is the estimation of voter transitions for all voters and there are differences between voters by mail and ballot box voters.

4.3.3 Population changes

The fourth scenario (II.4) covers population changes in the districts. All voters are clearly assignable to one district and there are no voters by mail. The data again originates from scenario I.3 (see Table 4). The first subscenario (II.4.1) only covers the changes because of generational change and naturalization. Around 4 % of the voters decrease between elections and 4 % are first time voters. To simulate a different balloting behavior, we draw the first from the high loyalty population group, while the first time voters are newly drawn to 75 % from the low loyalty group and to 25 % from the medium loyalty population group. Only the sensible election is included and differences in district populations are absorbed into the non-voters.

The subscenario II.4.2 and II.4.3 are based on II.4.1 and additionally accounts for population changes because of movers. We differentiate the movers between in town movers and external movers. On average, 25 % of all voters in a district are considered as in town movers. These movers are redistributed between all districts. External movers leaving the town also make up, on average, 25 % of the population. These voters are removed from the population for the second election. External movers coming to the town are, on average 35 % of the district. These voters are taken from the corresponding districts in scenario I.2 (see Table 4). Therefore, the new voters are still slightly similar in their voting behavior to the old population. The higher percentage of new external voters leads to higher populations in the second election. In II.4.2, population differences are covered using the proportional adjustment explained in Sect. 3.2. In subscenario II.4.3, the differences are absorbed into the non-voters.

We only use observable aggregated data for the estimation. Incorporating generational change (II.4.1) leads to no change in the structure of the estimates (see Fig. 3). The hierarchical model outperforms all other models. Surprisingly, the quality of the estimates is slightly better than in the underlying scenario I.3. Comparing the complete voter transition shows much worse estimates for scenario II.4.2 and II.4.3. The ADs of all models are around 0.6. One reason for the relative high figures is the design. At the simulated individual level, we have voters, who have only voted in one election. For the voter transition, we added the one-time voters to the non-voters of the other election. Due to this handling, the number of non-voters in our “true” individual data is much higher than in the aggregated data.

Therefore, the voter transition in the individual data differs. In Fig. 3 we present results using a different calculation: We restrict the comparison to the inner cells of the table, which contains all party transitions, but no longer the transitions involving the non-voters. Restricted to the inner cells, the models have estimates with a performance similar to the underlying scenario I.3. Absorbing the population changes in districts into the non-voters (II.4.3) slightly outperforms the proportional adjustment approach (II.4.2) in our simulation.

5 Discussion

We compare different approaches for estimation of voter transitions based on ecological inference. In the first step of the analysis (Sect. 3.1), we evaluate the models using real election data. We find non-negligible differences within the model estimates using different settings and also between the models. Many of our results are in accordance with those from other authors. The issue of out of admissible range estimates using the Goodman regression is well documented and has been one reason for the development of advanced models. The reference party dependency of the Thomsen model is at least known to practitioners. The problems of long Markov chains and strong in chain autocorrelation in the Bayesian estimation of hierarchical models are ubiquitous in Bayesian statistics. However, the widely discussed issue of the decision to use weighted or unweighted Goodman regressions has only a minor impact in our assessment. Furthermore, the model dependency in Andreadis and Chadjipadelis’s recursive algorithm (Sect. 2.4) is not unexpected, but it contradicts a claim by the authors (see Andreadis and Chadjipadelis 2009). The assessment of the hierarchical model proposed by Greiner and Quinn leads to technical difficulties and therefore is not included in the simulation-based assessment. However, it cannot be concluded that the observed difficulties indicate a general problem of the model, because these may be caused by specific characteristics of our application case.

We assess the performance of the approaches with an extensive simulation-based study (Sect. 3.2) and find large differences in the performance between the models in the four scenarios (see Table 4). If the constancy assumption for the voter transition over the districts holds, the Goodman regression shows the best performance. However, the hierarchical multinomial-Dirichlet model is superior to the other models in simulations with more realistic voter transition structures. This holds if the assumptions of the model are fulfilled, but also in the more realistic scenarios which do not comply with the distributional assumption of the model. The recursive models have

a similar performance only in one of the more realistic scenarios, but perform worse when compared to the hierarchical multinomial-Dirichlet model in our other scenarios. Overall, the Thomsen model clearly shows the worst performance and produces unreliable results.

Whereas our assessment is motivated by the application of ecological inference to estimate voter transitions in a metropolitan city like Munich, we are convinced that the insights provided can be partially generalized and are a valuable contribution to the general discussion. Previous assessments have often been limited to the 2×2 -case of ecological inference or to one specific model. The different designs of the studies are an additional complicating factor for a comparison between models. The performance and drawbacks of the classic Goodman regression are already widely discussed in preceding contributions ([Achen and Shively 1995](#); [King 1997](#)). [Thomsen \(1987\)](#) model has been rarely considered in other scientific study assessing ecological inference. Evaluations of the newer hierarchical multinomial-Dirichlet model and the recursive models are part of their initial publication ([Rosen et al. 2001](#); [Andreadis and Chadjipadelis 2009](#)). [Andreadis and Chadjipadelis](#) also include the hierarchical multinomial-Dirichlet model in their assessment. [Kellermann \(2011\)](#)'s comparison of ecological inference approaches for voter transition is limited to the Goodman Regression, [Andreadis and Chadjipadelis](#) recursive model and [Thomsen's](#) logit approach for ecological inference and is not simulation-based.

Our assessment confirms previous findings considering the Goodman regression. The general poor performance of [Thomsen's](#) model compared to other models is a new and important finding. The better performance of the hierarchical multinomial-Dirichlet model compared to the recursive models in our assessment contradicts [Andreadis and Chadjipadelis's](#) findings, which are based on possibly too few districts. The variation of performance across scenarios reveals that the estimates depend on the assumed underlying data structure. In view of the discussion following [King \(1997\)](#), the limited consequences of aggregation bias for the hierarchical multinomial-Dirichlet model are surprising.

We examine the consequences of data preprocessing in the third step of the assessment. Most models are affected by the level of aggregation. We can also demonstrate that the handling of voters by mail is important. Ignoring voters by mail is the worst solution and leads to a loss of estimation quality. We demonstrate that the voter transition between the parties can be estimated using ecological inference even if there are relevant changes in the population between the elections. However, we are not able to identify a clearly superior approach to handle these changes.

The consequences of data preprocessing are rarely discussed in ecological inference assessments. We are not aware of a study, which covers data preprocessing issues in ecological inference in a similarly extensive manner as our study. [Kellermann \(2011\)](#) covers some of the issues in his work, however he does not evaluate the consequences. We are convinced that an examination of data preprocessing is necessary and consider our assessment as valuable contribution. Even if our assessment concentrates on typical voter transition data preprocessing issues, our results should be partially transferable to other applications.

Overall, our assessment proves that the estimation of voter transition using ecological inference can be influenced by two crucial decisions: the selected model and

the applied data preprocessing strategy. Based on the results of our assessment, we recommend the usage of the hierarchical multinomial-Dirichlet model for ecological inference of voter transitions, if the number of districts is not too low. Furthermore, data preprocessing should be performed carefully. In case of voter transition, the number of available districts should be as high as possible and voters by mail should not be ignored. The former can be considered as general advice for ecological inference.

One problematic issue in ecological inference is the interval estimation, i.e. the calculation of confidence or credibility intervals. For the Bayesian approaches, they can be calculated from the resulting posterior distributions, but the performance of these intervals appeared to be weak. An issue for further research is to develop reliable credibility intervals for the hierarchical multinomial-Dirichlet model.

Furthermore, we use ecological inference methods only based on aggregate data without individual level data in our assessment. One extension of ecological inference (Wakefield 2004; Greiner and Quinn 2009) is the combination of both sources of information, aggregated and individual level data, to improve the overall performance. Such models are called “hybrid models”. The hierarchical multinomial-normal model proposed by Greiner and Quinn (2009) already allows the inclusion of individual data. For the recommended hierarchical multinomial-Dirichlet model a modified version, which allows the inclusion of individual level data, is available for R in the package “eiwild”. However, because (sufficient) individual level data will not be available in many cases, pure ecological inference will continue to be of high practical relevance.

Acknowledgments We would like to thank Uta Thien-Seitz and Sibel Aydemir for helpful discussions during the project. We are also grateful for the valuable comments of the anonymous referees.

6 Appendix

6.1 Model description Greiner and Quinn

As distributions at the district level for the Greiner and Quinn (2009) model several multinomial distributions are assumed:

$$((\text{CSU}_1, \text{CSU}_2)_i, (\text{CSU}_1, \text{SPD}_2)_i, \dots, (\text{CSU}_1, \text{NonVoters}_2)_i) \\ \sim \text{Multinomial}(\beta_{\text{CSU}, \text{CSU}, i}, \beta_{\text{CSU}, \text{SPD}, i}, \dots, \beta_{\text{CSU}, \text{NonVoters}, i}, \text{CSU}_{1,i}) \quad (14)$$

Equation (14) presents the assumed multinomial distribution for one row. Instead of the margins like in the Rosen et al. (2001) proposal, the inner cells, e.g. $(\text{CSU}_1, \text{CSU}_2)_i$, of the table are considered and assumed to be dependent from the parameters

$$\beta_{\text{CSU}, i} \sim (\beta_{\text{CSU}, \text{CSU}, i}, \beta_{\text{CSU}, \text{SPD}, i}, \dots, \beta_{\text{CSU}, \text{NonVoters}, i}) \quad (15)$$

and the corresponding row-margins $\text{CSU}_{2,i}$. For each of the R -rows, one independent multinomial distribution is assumed. To hold the restrictions defined by the margins, indicator functions for the row and column sums are used.

For the second level of the models, the parameters are logit transformed using a reference column (party). If the non-voters would be used as reference party, the transformed β -parameters from (14) would have the following form.

$$\begin{aligned}
\Omega_{\text{CSU},i} &= (\omega_{\text{CSU},\text{CSU},i}, \omega_{\text{CSU},\text{SPD},i}, \dots, \omega_{\text{CSU},\text{Others},i}) \\
&= (\log(\frac{\beta_{\text{CSU},\text{CSU},i}}{\beta_{\text{CSU},\text{NonVoters},i}}), \log(\frac{\beta_{\text{CSU},\text{SPD},i}}{\beta_{\text{CSU},\text{NonVoters},i}}), \dots, \log(\frac{\beta_{\text{CSU},\text{Others},i}}{\beta_{\text{CSU},\text{NonVoters},i}}))
\end{aligned} \quad (16)$$

The parameters of each multinomial distribution (row) are transformed. Therefore, the $R \times C$ β -parameters are reduced to $R \times (C - 1)$ ω -parameters. As common distribution for the district ω -parameters, a normal distribution is assumed at the second level.

$$(\Omega_{\text{CSU},i}, \Omega_{\text{SPD},i}, \dots, \Omega_{\text{NonVoters},i})^T \sim N_{R \times (C-1)}(\mu, \Sigma) \quad (17)$$

As hyperpriors for the μ a normal distribution and for the Σ an inverse wishart distribution are proposed by Greiner and Quinn (2009).

6.2 Description of the 2005 and 2009 Federal elections in Munich

See Fig. 4.

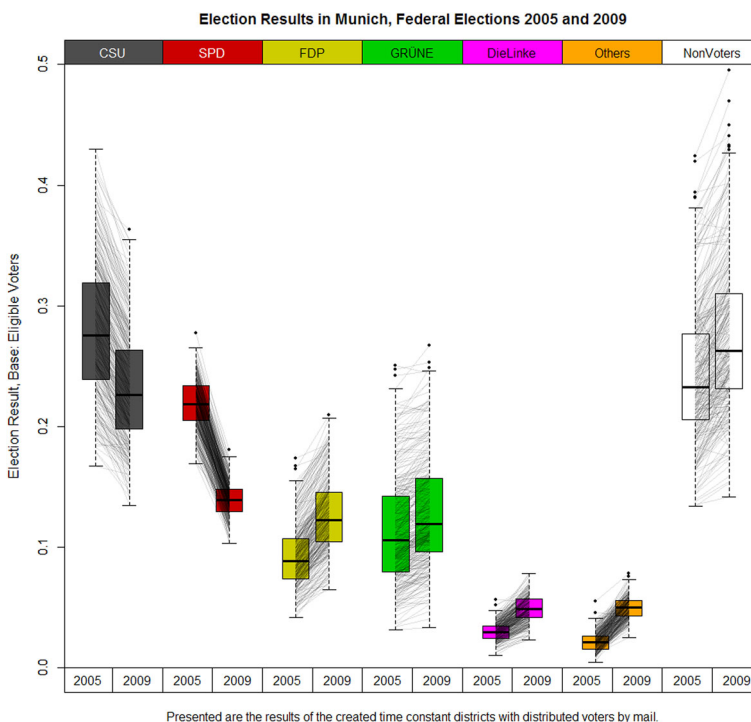


Fig. 4 Election results of the 2005 and 2009 Federal election in Munich. Results of the second vote (German: Zweitstimme) for the 314 time constant districts are presented. Voters by Mail are added to their corresponding ballot box districts. Each district is represented by *one line* for each party. For more details see Sect. 3.1

6.3 Additional model evaluation results

See Tables 9, 10, 11.

Table 9 Model evaluation Goodman, AD between model estimates is provided, unweighted [uw] or weighted [we] regression and two adjustment of estimates outside the admissible range, the truncation approach (3) [a] and our proposal (4) [o], are considered, for more details see Sect. 3.1

Model	uw,a	we,a	uw,o	we,o
uw,a	0.00	0.07	0.20	0.18
we,a	0.07	0.00	0.22	0.17
uw,o	0.20	0.22	0.00	0.10
we,o	0.18	0.17	0.10	0.00

Table 10 Model evaluation hierarchical Multinomial-Dirichlet model (eiPack), AD between Model estimates are provided, different values for burning, thinning and chain length (B/T/C) were used, for details see Sect. 3.1

Model	1000/ 1/ 1000	10,000/10/ 1000	10,000/100/ 2000	50,000/1000/1000	50,000/3000/1000
1000 /1 /1000	0.00	0.57	0.67	0.69	0.69
10,000/10 /1000	0.57	0.00	0.14	0.15	0.15
10,000/100 /2000	0.67	0.14	0.00	0.03	0.02
50,000/1000/1000	0.69	0.15	0.03	0.00	0.01
50,000/3000/1000	0.69	0.15	0.02	0.01	0.00

Table 11 Model evaluation Thomsen's logit approach, AD between model estimates are provided, different reference parties analyzed (identical for both elections); all other parameters are standard, for details see Sect. 3.1

Reference party	CSU	SPD	FDP	GRÜNE	DieLinke	Others	NonVoters
CSU	0.00	0.22	0.20	0.19	0.45	0.52	0.39
SPD	0.22	0.00	0.29	0.25	0.37	0.46	0.34
FDP	0.20	0.29	0.00	0.23	0.44	0.50	0.37
GRÜNE	0.19	0.25	0.23	0.00	0.40	0.44	0.32
DieLinke	0.45	0.37	0.44	0.40	0.00	0.32	0.21
Others	0.52	0.46	0.50	0.44	0.32	0.00	0.20
NonVoters	0.39	0.34	0.37	0.32	0.21	0.20	0.00

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